# Research Laboratories: System Design Description

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| Document number | SHV-RL-SDD-0001 |
| Revision | C |
| Status | APPROVED |
| Classification | INTERNAL |
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| Approved by | Lucan Merriweather |
| Date | 2024-11-12 |

## 1. System overview

The research laboratories host materials science, biology and astronomy groups: wet and dry labs, cleanroom suites, a zero-gravity laboratory and the station observatory.

The system is led by Science & Research. It is delivered through 6 work packages with a combined budget of 67,307.94 Mcr.

## 2. Components

| Code | Component | Specification | Drawing | Bill of materials |
|---|---|---|---|---|
| RL-MLB | Materials laboratory | SHV-RL-SPEC-0101 | SHV-RL-DWG-0201 | SHV-RL-BOM-0301 |
| RL-CLS | Cleanroom suite | SHV-RL-SPEC-0102 | SHV-RL-DWG-0202 | SHV-RL-BOM-0302 |
| RL-ZGL | Zero-gravity laboratory | SHV-RL-SPEC-0103 | SHV-RL-DWG-0203 | SHV-RL-BOM-0303 |
| RL-OBS | Observatory telescope | SHV-RL-SPEC-0104 | SHV-RL-DWG-0204 | SHV-RL-BOM-0304 |

## 3. Architecture

### 3.1 Materials laboratory

The materials laboratory is specified in SHV-RL-SPEC-0101. Its governing parameter is the bench count, set at 81 ea. The responsible engineer is Callum Radlow.

### 3.2 Cleanroom suite

The cleanroom suite is specified in SHV-RL-SPEC-0102. Its governing parameter is the cleanroom class, set at 5 class. The responsible engineer is Quinn Aster.

### 3.3 Zero-gravity laboratory

The zero-gravity laboratory is specified in SHV-RL-SPEC-0103. Its governing parameter is the lab volume, set at 5,025 m³. The responsible engineer is Talia Orlen.

### 3.4 Observatory telescope

The observatory telescope is specified in SHV-RL-SPEC-0104. Its governing parameter is the primary mirror diameter, set at 10.1 m. The responsible engineer is Darek Fairholm.

## 4. Interfaces

| Document | Title |
|---|---|
| SHV-ICD-PD-RL | Interface Control Document: Power Distribution / Research Laboratories |
| SHV-ICD-RL-CN | Interface Control Document: Research Laboratories / Computer Core & Data Network |

## 5. Work packages

| Work package | Name | Owner | Status | Complete |
|---|---|---|---|---|
| WP-SHV-RL-01 | Research Laboratories: Design & analysis | Saela Wade | IN_PROGRESS | 95.0% |
| WP-SHV-RL-02 | Research Laboratories: Fabricate & install materials laboratory | Vela Merrick | IN_PROGRESS | 81.4% |
| WP-SHV-RL-03 | Research Laboratories: Fabricate & install cleanroom suite | Saela Wade | IN_PROGRESS | 96.8% |
| WP-SHV-RL-04 | Research Laboratories: Fabricate & install zero-gravity laboratory | Quinn Aster | IN_PROGRESS | 75.5% |
| WP-SHV-RL-05 | Research Laboratories: Fabricate & install observatory telescope | Esme Crestwell | IN_PROGRESS | 79.0% |
| WP-SHV-RL-06 | Research Laboratories: Integration & test | Vela Merrick | IN_PROGRESS | 60.8% |

## 6. Risks

- Supply of materials laboratory assemblies depends on a single supplier.
- Supply of cleanroom suite assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.